"Racial Gaps in Managerial Career Advancement"

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ReadMe File

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Materials included

HPS Data preparation.R

This code builds the analysis data sets from the raw inputs in `data/`. It cleans and classifies job spells, compiles the race codings from the picture survey, matches every job spell to the league games played by that club during the spell, and constructs the career histories used in the simulations. It writes `data/analysis_environment.RData`, which holds every object the results code needs for the regression results. Run this first.

HPS Simulation preparation.R

This code fits the multinomial job-transition model and runs the simulations behind the counterfactual figures. It starts from `data/analysis_environment.RData` and adds nothing to it, so the preparation code has to have been run first. It writes one of `data/data_environment_all_<reps>.RData`, `_uk_<reps>.RData` or `_ms_<reps>.RData`, depending on the `uksim` and `reps` settings near the top of the script:

`uksim`	Sample	Writes
`0`	All former players	`data_environment_all_<reps>.RData`
`1`	UK nationals only	`data_environment_uk_<reps>.RData`
`2`	Propensity-score matched	`data_environment_ms_<reps>.RData`

The replication count is stamped into the file name, so a short test run writes its own files rather than overwriting the full ones. `HPS Simulation results.R` has a matching `reps` setting that selects which run to report on; it defaults to 1000, the environments that ship here.

The results code loads all three, so this script has to be run three times, once per setting. Only the simulation sample changes between runs, so `analysis_environment.RData` does not need rebuilding in between. Each run fits the 1000-replication simulation and takes several hours. The three environments are included in `data/`, so the results code can be run without repeating them.

HPS Racial Gaps.R

This code generates every result in the paper and the online appendix apart from those built on the simulations. Sections are marked with headers so they can be collapsed in the RStudio side-bar to give an overview of all tables and figures. After running the preamble, which loads `analysis_environment.RData`, each chunk can be run independently to obtain the results for a given table or figure. Output is written to `output/`.

HPS Simulation results.R

This code generates the results built on the simulations: the counterfactual prevalence figures, the headline decomposition in Table A.5, and the multinomial transition model in Appendix Table D.1. It loads the three simulation environments, so `HPS Simulation preparation.R` has to have been run for each `uksim` setting first. Output is written to `output/` alongside the rest.

All four scripts resolve paths relative to the package folder, so set the working directory to the package root before running:

```
setwd("/path/to/Replication Package")
source("HPS Data preparation.R")
source("HPS Simulation preparation.R")   # once per uksim setting
source("HPS Racial Gaps.R")
source("HPS Simulation results.R")
```

Required packages: `collapse`, `plyr`, `dplyr`, `tidyverse`, `ggplot2`, `lfe`, `xtable`, `RColorBrewer`, `survival`, `Matching`, `nnet`, `data.table`, `stringr`. `tidyverse` is needed for `tidyr` and `purrr`.

data/ and **output/**

`data/source/` holds the raw inputs; `data/` itself holds the environments built from them. `output/` holds what the results code writes, and is overwritten each time it runs. It is split three ways:

Folder	Holds
`output/tables/`	the 13 tables in the paper, as `.tex`
`output/figures/`	the 26 numbered figure files the code produces, as `.pdf`
`output/other/`	5 figures the code produces that the paper does not report

Output files are named for the table or figure they are in the paper, so `tables/table4_hazard_model.tex` is Table 4 and `figures/figure6_survival_2x2.pdf` is Figure 6. Where one numbered exhibit is built from several panels, the panels share the number and differ by suffix.

Paper	File	Written by
`Table 1`	`tables/table1_summary_stats_players.tex`	HPS Racial Gaps.R
`Table 2`	`tables/table2_summary_stats_management.tex`	HPS Racial Gaps.R
`Table 3`	`tables/table3_entry_regs.tex`	HPS Racial Gaps.R
`Table 4`	`tables/table4_hazard_model.tex`	HPS Racial Gaps.R
`Table A.1`	`tables/tableA1_job_descriptions.tex`	HPS Racial Gaps.R
`Table A.2`	`tables/tableA2_coaching_licence.tex`	HPS Racial Gaps.R
`Table A.3`	`tables/tableA3_entry_regs_logit.tex`	HPS Racial Gaps.R

Paper	File	Written by
Table A.4	tables/tableA4_spell_coverage.tex	HPS Racial Gaps.R
Table A.5	tables/tableA5_headline_decomposition_<reps>.tex	HPS Simulation results.R
Table B.1	tables/tableB1_covariate_balance.tex	HPS Racial Gaps.R
Table B.2	tables/tableB2_entry_regs_matched.tex	HPS Racial Gaps.R
Table C.1	tables/tableC1_entry_regs_uk.tex	HPS Racial Gaps.R
Table D.1	tables/tableD1_simplified_transition_model_<reps>.tex	HPS Simulation results.R
Figure 3	figures/figure3_days_as_manager_{eng,row}_{all,uk}.pdf	HPS Racial Gaps.R
Figure 4	figures/figure4_career_progression_{all,ukonly}.pdf	HPS Racial Gaps.R
Figure 5	figures/figure5_job_transitions.pdf	HPS Racial Gaps.R
Figure 6	figures/figure6_survival_2x2.pdf	HPS Racial Gaps.R
Figure 7	figures/figure7_manager_sim_vs_actual_{eng,row}_<reps>.pdf	HPS Simulation results.R
Figure 8	figures/figure8_manager_sim_<reps>.pdf	HPS Simulation results.R
Figure 9	figures/figure9_assman_sim_<reps>.pdf	HPS Simulation results.R
Figure A.1	figures/figureA1_job_transitions_unrestricted.pdf	HPS Racial Gaps.R
Figure A.2	figures/figureA2_employment_sim_vs_actual_<reps>.pdf	HPS Simulation results.R

Paper	File	Written by
Figure A.3	`figures/figureA3_youth_sim_<reps>.pdf`	HPS Simulation results.R
Figure B.1	`figures/figureB1_propensity_scores.pdf`	HPS Racial Gaps.R
Figure B.2	`figures/figureB2_days_as_manager_{eng,row}_ms.pdf`	HPS Racial Gaps.R
Figure B.3	`figures/figureB3_manager_sim_vs_actual_{eng,row}_ms_<reps>.pdf`	HPS Simulation results.R
Figure B.4	`figures/figureB4_manager_sim_ms_<reps>.pdf`	HPS Simulation results.R
Figure B.5	`figures/figureB5_assman_sim_ms_<reps>.pdf`	HPS Simulation results.R
Figure C.1	`figures/figureC1_manager_sim_vs_actual_{eng,row}_uk_<reps>.pdf`	HPS Simulation results.R
Figure C.2	`figures/figureC2_manager_sim_uk_<reps>.pdf`	HPS Simulation results.R
Figure C.3	`figures/figureC3_assman_sim_uk_<reps>.pdf`	HPS Simulation results.R

The paper references 27 figure files in total. Figure 2 is the twenty-seventh and is not produced by this code: it is a screenshot of the picture survey, `survey.png`, which has to be supplied separately. Figure 1 has no file at all, being a drawn diagram of the career pipeline.

Files written by `HPS Simulation results.R` carry the replication count of the environments they were built from, so `<reps>` is `1000` for the output that ships here. The files written by `HPS Racial Gaps.R` do not, since no simulation goes into them.

The five files in `output/other/` carry no number. They are the matched and UK-sample counterparts of figures the paper reports for one sample only, kept because the simulation produces all three samples anyway: `barplotday_jobcat_nomods_ms.pdf`, `employment_sim_vs_actual_{uk,ms}_<reps>.pdf` and `youth_sim_{uk,ms}_<reps>.pdf`.

Data collection and compilation

www.transfermarkt.com

Transfermarkt supplies the playing careers, the coaching careers and the club and competition structure. Player pages give season-by-season appearances, goals, assists and minutes by competition, along with date of birth, position and nationality. Coach pages give one entry per job spell, with the club, the job title, the start and end date, and the number of matches taken charge of. Club pages give the league, tier and final position for each season. All are collected with `rvest` in R, looping over profile URLs and parsing the relevant tables.

Match results

League results come from two sources. `allgames.Rdata` is a scrape of full league fixture lists covering 23 competitions in nine countries. It was assembled for Hoey, S. and K. Narita (2026), "The Legacy of Leadership: Role Modelling and Managerial Skill Transmission", working paper (available here), and is reused here rather than rebuilt. It leaves gaps: competitions outside those nine countries, tiers below the top one or two, and seasons in which a club dropped out of a covered division. To fill them we run a targeted scrape of club fixture pages, looping over job spells whose games are not already covered, using

```
paste0("https://www.transfermarkt.com/x/spielplan/verein/", clubid, "/saison_id/", season)
```

where `clubid` is the club's Transfermarkt identifier and `season` is the calendar year in which the season begins. This produces `gamedata_scraped.Rdata`. Scores on those pages are recorded in true home-away orientation rather than from the club's perspective, and dates are disambiguated using the printed weekday. The two sources overlap, so they are combined on the numeric Transfermarkt match identifier rather than by row, which removes roughly eleven thousand duplicated games.

Race coding

Race is coded from player photographs using a purpose-built picture survey. Respondents were shown a photograph and asked to classify the player. Each player was rated by several respondents; `responsesperid` aggregates these into vote shares and a majority classification. Players without a clear majority are flagged by `nomaj` and excluded from the analysis.

Compilation

Job spells are read from the coaching data, cleaned of duplicate scrapes, and classified into manager, assistant manager, youth team management and other staff using the job title. Youth and reserve sides are identified from the club identifier and separated from senior roles. Each spell is assigned a country from the club, and a unique spell identifier, since the same person can hold two roles at one club from the same start date. Spells are then matched to games: a game counts toward a spell when it was played by that club between the spell's start and end dates. From the matched games we build cumulative points, a rolling ten-game performance measure, and cumulative experience. A spell enters the duration analysis when at least ten of its games are matched, which is the minimum needed for the rolling measure.

We anonymize player and coach names, club names and profile URLs by applying a number to each unique value. Only the year of birth is retained, since the specifications use year-of-birth fixed effects.

Part I. Analysis data frames

These are the data frames the results code uses. They are produced by `HPS Data preparation.R`, and the simulation environments by `HPS Simulation preparation.R`. All of them are included in `data/` so that the results code can be run without rebuilding them.

analysis_environment.RData

Loaded in the preamble of `HPS Racial Gaps.R`, and again by `HPS Simulation results.R` for the plotting constants and helper functions. Holds every object behind the results in the paper apart from the simulations.

`regframe`

The estimation sample for the duration analysis. The unit of observation is **a game within a job spell**: one row for every league game played by the club while the person held the job. Used for the Cox hazard models and the survival curves.

Variable	Description
`jobspellid`	Unique job spell identifier
`link`	Anonymized person identifier
`dob`	Date of birth, year only
`club`	Anonymized club name
`country`	Country of the club
`funccat`	Job category: Manager, Assistant manager, Youth team management/development, Other staff
`funccat_alt`	Job category, keeping non-England spells separate rather than pooling them
`funccat_bn`	Job category split by location, e.g. Manager England, Manager ROW
`youth`	Dummy for a youth or reserve side
`coachlicence`	Highest coaching licence held
`licencedum`	Dummy for holding any coaching licence
`startdatenum`	Spell start date, numeric
`enddatenum`	Spell end date, numeric; open-ended spells take the value 19723
`date` , `datenum`	Date of the game
`div`	Competition in which the game was played
`season`	Season in which the game was played
`sc` , `op_sc`	Goals scored by the club and by the opponent
`points`	Points won in the game (win = 3, draw = 1, loss = 0)
`cumulpoints`	Cumulative points won during the spell
`cumulgames`	Games elapsed in the spell; the duration model's time variable
`cumulpoints_last10`	Points won over the last ten games; the performance control
`cumulgamestot`	Cumulative games worked across the whole coaching career

Variable	Description
`cumulgamesstart`	Games worked as an (assistant) manager before this spell began
`leftjob`	Dummy set to 1 on the last game of a spell that ended; the duration model's event
`engspell`	Dummy for the spell being at an English club
`black`	Dummy for the coach being Black, from the majority survey classification
`nomaj`	Dummy for no clear survey majority; these observations are excluded
`position`	Playing position
`sum_appearances_tier1` … `_lowtier`	Career appearances by tier of English football
`sum_goals_tier1` … `_lowtier`	Career goals by tier
`sum_assists_tier1` … `_lowtier`	Career assists by tier
`sum_minplayed_tier1` … `_lowtier`	Career minutes played by tier
`nationality` , `nationality2`	Player nationalities
`natcat` , `natcat2`	Nationality grouped into UK, Europe, Africa, ROW
`ukonly`	Dummy for holding a UK nationality and no other
`ukplus`	Dummy for holding a UK nationality alongside another
`foreign`	Dummy for holding no UK nationality; UK-only is the omitted category

`coachdata`

One row per **job spell**. Used for the summary statistics, the job-category counts, and to derive the England-located job indicators in the entry regressions.

Variable	Description
`link` , `dob`	Anonymized person identifier and year of birth
`club`	Anonymized club name
`clubfunc` , `func`	Job title as recorded
`funccat` , `funccat_alt` , `funccat_bn`	Job categories, as in `regframe`
`start` , `end`	Spell start and end as recorded, season and date
`startdatenum` , `enddatenum`	Spell start and end, numeric
`country`	Country of the club
`youth`	Dummy for a youth or reserve side
`coachlicence` , `licencedum`	Coaching qualification held, and a dummy for holding any
`min.season` , `max.season`	First and last season of the person's playing career
`jobdone` , `jobdone_alt` , `jobdone_bn`	Dummies for having held each job category

`playerlevel`

One row per **player**, summarising the playing career. The basis for `playerlevelstats`, which is the estimation sample for the entry regressions and the summary statistics.

Career totals by tier (`sum_appearances_*`, `sum_goals_*`, `sum_assists_*`, `sum_minplayed_*`), per-game averages (`avg_goalspergame_*`, `avg_assistspergame_*`, `avg_minplayedpergame_*`), playing position (`position`, `attacker`, `midfielder`, `defender`, `goalkeeper`), nationality (`nationality`, `nationality2`, `natcat`, `natcat2`, `ukonly`, `ukplus`, `foreign`, `natteam`, `natapps`), date of birth, and dummies for each job category ever held (`Manager`, `Manager_England`, `Manager_ROW`, `Assistant_manager`, `Assistant_manager_England`, `Youth_England`, and so on) which become the outcomes in the entry regressions.

`jobevol`

Career histories. The unit of observation is a **job spell with the preceding and following spell attached**. Used for the job-transition figures.

Variable	Description
`link`	Anonymized person identifier
`funccat`, `funccat_alt`, `funccat_bn`	Job category of the current spell
`previous_funccat`, `previous_club`, `previous_country`, `previous_endspell`	The preceding spell
`next_funccat`, `next_club`, `next_country`, `next_startspell`	The following spell
`startdatenum`, `enddatenum`	Spell start and end, numeric
`days_elapsed`	Days since the end of the playing career
`state`, `state_alt`, `state_bn`	Job state at this point in the career; the outcome in the transition model
`tier_end`, `tier_next`	Tier of the club at the end of this spell and at the start of the next
`totaljobs`	Number of jobs held over the career
`yob`	Year of birth
`natcat`	Nationality group

`responsesperid`

Race coding aggregated to the player. One row per **player rated in the survey**. Supplies the `black` and `nomaj` variables used throughout.

Variable	Description
`id`	Survey photo identifier
`link`	Anonymized person identifier
`black`, `other`, `dn`	Respondents classifying the player as Black, as another background, or don't know
`total`	Number of respondents
`blackshare`, `othershare`	Vote shares

Variable	Description
`blackmaj` , `othermaj`	Dummies for a majority classification
`nomaj`	Dummy for no clear majority; these players are excluded
`disagree`	Dummy for disagreement among respondents

`tiercount`

One row per player-tier. Counts of seasons played in each tier of English football, merged into `playerlevelstats` for the summary statistics.

data_environment_all_1000.RData, _uk_1000.RData, _ms_1000.RData

Loaded by `HPS Simulation results.R` only, one per sample: `_all` for all former players, `_uk` for UK nationals, `_ms` for the propensity-score matched subsample. The trailing number is the replication count, and the `reps` setting in that script picks which one to load. Each contains the output of a 1000-replication simulation. They are trimmed to the objects the simulation code actually reads; the regression results all come from `analysis_environment.RData` instead.

All three hold the five `plotframe_share` series and `statelist_simact`. Only `_all` also holds `date_sequence_sim`, which is read by Appendix Table D.1 alone; carrying it in `_uk` and `_ms` would add several hundred megabytes in memory for nothing.

`date_sequence_sim`

In `data_environment_all_1000.RData` only. The career panel the transition model is estimated on. The unit of observation is a **player in a 14-day interval** since the end of their playing career.

Variable	Description
`link`	Anonymized person identifier
`dob` , `yob`	Year of birth
`days_elapsed`	Days since the end of the playing career
`black`	Dummy for the person being Black
`funccat_bn`	Job state in this interval
`funccat_bn_lag`	Job state in the previous interval; the model's lagged state
`Manager`	Dummy for holding a manager job
`position` , `attacker` , `midfielder` , `defender` , `goalkeeper`	Playing position
`nationality` , `nationality2` , `nationality3`	Player nationalities
`natteam` , `natapps`	National-team experience and appearances
`uk` , `UK` , `ROW` , `ukonly` , `ukplus` , `foreign`	Nationality indicators
`sum_goals_*` , `sum_assists_*` , `sum_minplayed_*`	Career playing statistics by tier

`plotframe_share, plotframe_share_simact, _simeao, _simpao, _simom`

Prevalence series used for the simulation figures and the counterfactual decomposition. One row per **day, race and job state**.

Variable	Description
`days_elapsed`	Days since the end of the playing career
`black`	Race group
`funccat_bn`	Job state
`n`	Number of people in that state
`share`	Share of the race group in that state
`type`	Scenario label

The five frames are the actual series (`plotframe_share`), the simulated actual series (`_simact`), and the three counterfactuals: entry as the other race (`_simeao`), progression as the other race (`_simpao`), and both (`_simom`).

`statelist_simact`

A list of **1000 elements**, one per simulation replication, each a data frame with the same columns as `plotframe_share` . Used to compute the 2.5th and 97.5th percentile bands shown on the simulation figures.

Part II. Source data frames

These are the raw inputs. They are read by `HPS Data preparation.R` only, and are not used by the results code.

File	Object	Unit of observation	Contents
`playercareerlevel250423.Rdata`	`playercareerlevel`	Player-season-club	Appearances, goals, assists and minutes, by season and competition
`coachdata030523.Rdata`	`coachdata`	Job spell	Club, job title, start and end date, coaching licence
`clubdata250423.Rdata`	`clubdata`	Club-season	League, tier, wins, draws, losses, goals, points, final rank, country
`natframe080523.Rdata`	`natframe`	Player	Nationalities, up to two per player
`leaguedata300623.Rdata`	`leaguedata`	Competition	Competition identifier and the country it belongs to
`combinedresponses.Rdata`	`combined_responses`	Survey response	One row per respondent per photograph: the classification given, plus respondent demographics
`playerids.Rdata`	`ids`	Player	Link between the survey photo identifier and the player
`all_clublinks_countries.csv`	`countries_app`	Club	Club to country mapping
`jobdescrip.csv`	`jobdescrip`	Job title	Job title to job category mapping
`allgames.Rdata`	`gameframes`	Team-match	League match results for 23 competitions in nine countries: date, competition, season, teams, score. One row per team per match, 595,830 rows. From Hoey and Narita (2026), see Data collection and compilation
`gamedata_scraped.Rdata`	`gamedata_scraped`	Team-match	Targeted fixture scrape covering competitions and seasons absent from `allgames`

`allgames.Rdata` and `gamedata_scraped.Rdata` are reduced to the columns the preparation code reads. The original scrape of `allgames` is at player-appearance level and carries team sheets, attendances and referees, none of which the analysis uses; it is collapsed to one row per team per match before distribution, which is the first thing the preparation code does with it in any case.

`durationdata.Rdata` holds `masterdf`, the intermediate join of job spells to games. It is written and re-read by the preparation code and is not used by the results code.